CHAPTER 19

Principles of Nuclear Power

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From earliest times man has become increasingly dependent of a variety of energy sources: heat from burning of wood, animal dung, coal, oil, natural gas, etc., mechanical energy from steam engines, wind mills and water falls, and electricity from the same sources as well as from nuclear reactors. Much effort has been invested in searches for new energy sources, preferably of the "renewable" kind in contrast to the non-renewable sources[1]. Energy is a strategic resource, playing an important role in international and

[1] In a wider context, all energy originates from gravitational collapse, annihilation, fusion reactions or mass loss into black holes and its sources are thus never renewable.

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national politics. This chapter will describe the principles of current nuclear energy generation by fission and also briefly discuss the possibility to use fusion for the same purpose.

After the discovery by Hahn, Strassman, Meitner and Frisch in 1938–1939 that neutrons induced fission in uranium, and that the number of neutrons released in fission was greater than one, many scientists realized that it should be possible to build a chain-reacting system in which large amounts of nuclear energy were released, *a nuclear reactor.* The first such system was constructed in Chicago in the early 1940s under the scientific leadership of Fermi and became critical on December 2, 1942, as part of the World War II Manhattan Project. Since that time many hundreds of nuclear reactors have been built throughout the world, mostly for power production.

In mid 2012, 433 nuclear power reactors (372 GW_e)[2] were in operation; 122 (114 GW_e) in North America including the US, 6 (~4 GW_e) in South America, 2 (~2 GW_e) in Africa, 186 (~162 GW_e) in Europe, including the Russian federation, and 119 (~88 GW_e) in the Far East (mainly Japan). Presently 62 power reactors (~59 GW_e) are under construction, the majority in the Far East and most of the rest in Europe. As small old nuclear power plants are shut down and replaced by bigger new ones, the number of reactors may remain constant in a geographic area, or even decrease slightly, although the total installed nuclear generating capacity often continues to increase.

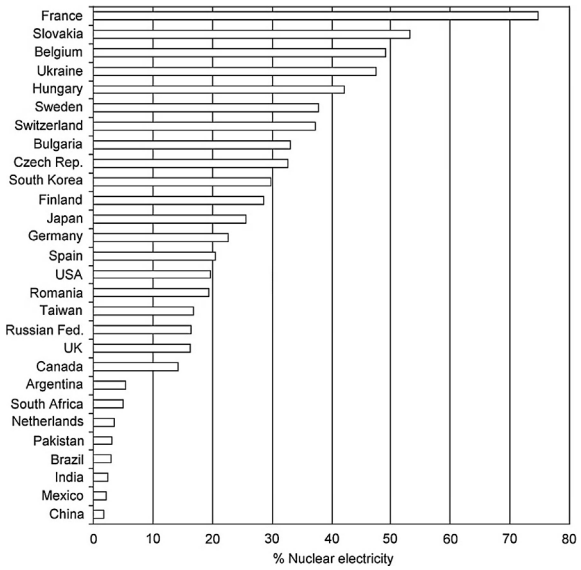


Figure 19.1 Relative amount of all electricity in each country generated by nuclear power stations during 2010.

[2] GW_e = gigawatt electric power installed.

Figure 19.1 shows the fraction of all electricity generated by nuclear power stations in different countries. In France any further large increase in nuclear power production capacity is unlikely as this could lead to a generating capacity in excess of demand. Furthermore, opposition to the peaceful use of nuclear power has led to revised plans for the use of nuclear power in many countries in the West (cf. Ch. 22). In some cases earlier expansion plans have been replaced by moratoria on new plants, by a decision to shut-down some or all operating nuclear power stations (Germany, Italy, and Lithuania), or by plans to force the utilities to shut down all nuclear power stations at some future date (Germany).

Nuclear chemistry plays an essential part in achieving safety and reliability in this source of power. Nuclear chemists and engineers are responsible for much of the nuclear fuel cycle, from uranium ore processing to ultimate disposal of radioactive waste.

In the future fusion power may also contribute to the energy supply. However, not all problems have been overcome to date, see §19.17.

19.1. THE NUCLEAR REACTOR

The fission process with thermal neutrons can be summarized as follows

$$^{235}\mathrm{U} + \mathrm{n} \rightarrow \mathrm{FP} + \nu\mathrm{n} \tag{19.1}$$

For each neutron consumed on the average ~ 2.5 ($= \nu$) new neutrons are released. The new neutrons can be used to fission other ^{235}U nuclei leading to the release of even more neutrons forming a *nuclear chain reacting system*. The kinetic energy of the fission fragments is promptly converted to heat through collisions with other atoms. In a *nuclear reactor* the nuclear chain reactions are controlled so that an equilibrium state is reached, where for each fission exactly one of the new neutrons is used for further fission. Under these conditions the *neutron multiplication factor k* is said to be 1. If the factor is higher the number of neutrons, and consequently the fission rate, increases exponentially. Without any control mechanism the heat evolved would ultimately destroy the chain-reacting system. This is extremely improbable in conventional reactors because of various active and passive control mechanisms. No such control mechanisms are present in *nuclear explosive devices* and, instead, every effort is made to make the reaction as violent as possible.

The neutron transport equation describes the interaction of neutrons with their environment. The accurate design of a chain reacting system requires solution of this equation. The complete solution yields the neutron flux and neutron spectrum as function of the space coordinates and time. As there is no known analytical solution to this equation for a real chain reacting system, numerical methods or various approximations must be used. In this text we do not consider numerical solution methods. We illustrate the general properties of chain reacting systems of interest to the nuclear